A Small Language Model Does Not Reliably Exhibit Classic Decision Biases: A Preregistered Null Result in Qwen3-1.7B

Anonymous CogSci submission

Abstract

Language models are increasingly used as stand-ins for human decision-makers, and several studies report that they exhibit human-like cognitive biases. We test the boundary of that claim with a preregistered pilot battery of five classic biases (framing, conjunction fallacy, anchoring, endowment effect, cognitive reflection), each probed with three items (framing uses three paraphrase wordings; the other biases use three paradigm instantiations), run on Qwen3-1.7B-Instruct under CPU-only inference. Our preregistered gate required at least three of five biases to show the human-direction effect in at least two of three items (one-sided tests, $n = 10$ per cell, $\alpha = 0.05$). The gate failed: only framing and anchoring passed (2/5). The failures are structural, not marginal: the model essentially never commits the conjunction fallacy (violation rate 0.00–0.20 against a human target of 0.85), produces the reverse of the human willingness-to-accept/willingness-to-pay ordering (median WTA below median WTP), and cognitive reflection is item-dependent (it matches the human aggregate magnitude yet passes only one of three classic items). We then tested whether inference-only "irrationality levers" (act-humanlike, gut, and descriptive-norm prompts) could calibrate the model's bias magnitudes to published human effect sizes. Every lever *increased* miscalibration relative to the neutral baseline (mean absolute error 0.358 for the baseline vs. 0.416–0.437 for the levers): there is nothing to calibrate when the biases are absent. Per the preregistered stopping rule, the planned held-out confirmatory test was not run. This is a null result about one small model under fixed protocols—not a claim about language models in general—and we discuss scale and elicitation as the two central limitations. All code, prompts, and a one-command reproduction package are provided as anonymized supplementary material for review.

Keywords: cognitive biases; language models; human simulation; null results; preregistration; decision-making

Introduction

Language models are being deployed as stand-ins for human participants: simulating survey respondents, economic agents, and experimental subjects Aher et al. (2023); Argyle et al. (2023). For this practice to be valid, models must reproduce not only what people *do on average* but the systematic ways in which human judgment deviates from rationality. That bar is currently not met. Liu et al. (2025) show that frontier LLMs, when asked to predict or simulate human choices on the 13,006 risky-choice problems of choices13k Peterson et al. (2021), correlate far more strongly with the rational maximum-expected-value model than humans do: models assume people are more rational than we really are. Aher et al. (2023) document a complementary "hyper-accuracy distortion": aligned models produce unnaturally precise, idealized responses that depart from typical human patterns.

The literature is split on where human-like bias in LLMs comes from and where it goes. Hagendorff et al. (2023) find that intuitive reasoning biases *emerge* with scale in base models but *disappear* in instruction-tuned ChatGPT variants; Itzhak et al. (2024) find the opposite for a different bias set, with the decoy effect, certainty effect, and belief bias *strengthening* after instruction tuning. Macmillan-Scott & Musolesi (2024) benchmark multiple LLM families on classic cognitive-psychology tasks and report broad but uneven susceptibility to irrational reasoning across tasks and model families. Two recent lines complicate any simple story: Pagliaro (2026) shows measured bias magnitudes carry large systematic uncertainty from prompt reformulation alone, and Knipper et al. (2025) shows bias susceptibility depends jointly on model size and prompt specificity. Meanwhile, inference-only levers clearly move behavior: declaring that a model should "act humanlike" increases measured bias Mohr (2025), and socioeconomic personas induce large, prospect-theory-consistent shifts in risk-taking Dubedy (2026).

Against this background, the strongest existing approach to human-like simulation is fine-tuning: Binz et al. (2025) fine-tune a language model on Psych-101 (10M+ human choices) to create CENTAUR, which predicts held-out human behavior better than cognitive models. Fine-tuning is powerful but expensive, data-hungry, and out of reach for CPU-only settings. We ask a narrower, falsifiable question: *without any weight updates*, can we calibrate a small model's *measured bias magnitudes* to match published human effect sizes, and does that calibration transfer to better reproduction of real human choice distributions on held-out items?

Our contributions: (1) a preregistered pilot bias battery with a gate criterion, run on Qwen3-1.7B-Instruct under CPU-only inference, whose failure we report honestly: only 2 of 5 classic biases show the human-direction effect, so the planned confirmatory test was stopped per the preregistered rule; (2) a calibration analysis showing that inference-only "irrationality levers" systematically *worsen* fit to human effect sizes rather than improving it, with the neutral baseline as the best-calibrated condition; (3) mechanism-level detail on where the biases break down (absent conjunction fallacy, reversed WTA/WTP ordering, item-dependent reflection), which reframes the null as a boundary condition for the literature

claiming human-like bias in language models; (4) a public one-command reproduction package, provided as anonymized supplementary material for review. We scope every claim to the tested model and protocols: this is evidence about Qwen3-1.7B under fixed elicitation, not about language models in general.

Related Work

Cognitive biases in LLMs. Early work established that LLMs display human-like biases across judgment tasks Hagendorff et al. (2023); Macmillan-Scott & Musolesi (2024); Echterhoff et al. (2024). The role of post-training is contested: biases vanish under instruction tuning in Hagendorff et al. (2023) yet emerge more strongly in Itzhak et al. (2024), suggesting bias expression is bias-, model-, and measurement-dependent rather than a monotone function of alignment.

LLMs as (over-)rational human models. Liu et al. (2025) compare LLM predictions and simulations against choices13k and find models align with expected-value maximization far more than humans do. Related 2025 work reports that prospect-theoretic modeling of LLMs is not consistently reliable across models and not robust to epistemic (linguistic) uncertainty, even when numeric risk tests pass Wang et al. (2025). Our work takes the over-rationality gap as its starting point.

Prompt sensitivity and measurement. Pagliaro (2026) introduce a Bias Strength Index with confidence intervals, decomposing uncertainty into statistical and systematic (prompt-reformulation) components across eleven biases and eight models; Knipper et al. (2025) assess bias across TELeR prompt-specificity levels, finding larger models (>32B parameters) reduce bias in 39.5% of cases while higher prompt detail reduces most biases by up to 14.9%. We adopt their lesson directly: every bias is probed in multiple formulations, and instability is reported, not hidden.

Persona and induction levers. Dubedy (2026) shows socioeconomic personas shift simulated gambling behavior with large effect sizes (Cohen's $d = 4.15$); a systematic thesis study finds "act humanlike" declarations increase measured bias across models and scenarios Mohr (2025). We used such levers as calibration instruments, selecting among them by fit to human effect sizes rather than by intuition; Section 4.2 reports that every lever worsened fit.

Fine-tuning counterpoint. Binz et al. (2025)'s CENTAUR fine-tunes on 10M human decisions and generalizes to new tasks and domains. We disclose it explicitly as the closest concurrent approach and the stronger one: our contribution is the *cheap, inference-only* complement, with preregistered falsification, for settings where fine-tuning is infeasible.

Concurrent prompt-bias calibration. A 2026 study manipulates "prompting bias" (strict-to-lenient instructions) in GPT-5 to approximate individual participants' evaluation tendencies on idea ratings, finding aligned settings improve human-likeness over neutral prompts Furukawa (2026). Our work differs in domain (cognitive biases in risky choice vs. idea evaluation), model scale (1.7B open vs. frontier reasoning model), calibration target (published human effect sizes vs. individual tendencies), and preregistered falsifiable design; we disclose it as the nearest concurrent attempt at the same underlying idea. A caution comes from Feng et al. (2026): persona steering can redistribute rather than reduce bias, operating in the output channel with limited structural reach—a limit our Discussion returns to.

Method

Model and inference

We use Qwen3-1.7B-Instruct (GGUF Q4_K_M quantization; `lm-kit/qwen-3-1.7b-instruct-gguf` on Hugging Face), run CPU-only via llama.cpp (2 threads). Qwen3 defaults to thinking mode, so every user message ends with `/no_think`, keeping generations short and parseable. Decoding: temperature 0.7, top-p 0.9, max 64 tokens; all sampling seeds are fixed per (problem, sample), so every run is exactly reproducible. No weight updates are performed anywhere.

Human ground truth

choices13k Peterson et al. (2021) contains 13,006 binary risky-choice problems with aggregate human choice rates (`bRate`, fraction choosing Gamble B, $n \approx 15$–33 participants per problem). Our analysis uses the 2,380 Block-1 (no-feedback, description-only) rows, matching the description paradigm presented to the model. Gamble order is randomized per sample with a seeded RNG to control primacy effects Echterhoff et al. (2024), and model letter choices are mapped back to the dataset's A/B labels.

Pilot bias battery and gate

Before any calibration, we run a five-bias battery on the *uncalibrated* model, three prompt formulations per bias (following Pagliaro (2026) on prompt instability): framing (Asian disease and paraphrases; human target: 72% vs. 22% certain-choice rate, i.e. 0.50 reversal), conjunction fallacy (Linda and isomorphs; target 0.85 violation), anchoring (low/high anchors; target 0.40 on the anchoring index, the mean estimate shift divided by the anchor span), endowment (WTA/WTP elicitation; target ratio 2.0), and CRT (three classic items; target 0.60 intuitive-error rate). Each cell uses $n = 10$ samples. The preregistered gate requires: (i) ≥3 of 5 biases show the human-direction effect in ≥2/3 formulations (one-sided $\alpha = 0.05$); (ii) ≥90% parse rate; (iii) no ceiling saturation. Per-item tests are one-sided with the no-bias null: two-proportion z (framing), exact binomial (conjunction, CRT), and Mann–Whitney normal approximation (anchoring, endowment), implementing the preregistered one-sided design with standard tests (documented as D5 in DEVIATIONS.md). Fifteen one-sided tests run uncorrected at $\alpha = 0.05$; for a null result this leniency cuts *against* the authors, and the gate outcome is unchanged under per-bias Bonferroni correction (Section 4.1). If the gate fails, the study stops and the negative result is reported.

Irrationality levers and calibration

Four fixed inference-only levers: **L0** neutral ("You are a helpful assistant. Answer with your choice."); **L1** act-humanlike (respond with ordinary human judgment, instincts, imperfections); **L2** gut/System-1 (first instinct, do not calculate); **L3** L1 plus a descriptive norm (most people deviate from expected-value maximization). For each bias b and lever L, calibration error is $e(b, L) = |\text{model_bias}(b, L) - \text{human_target}(b)|$ on a $[0, 1]$ scale. We select $L^* = \arg\min_L \frac{1}{5} \sum_b e(b, L)$ once, on the pilot battery only, then freeze it. L0 is the uncalibrated baseline. Uncertainty on the per-lever mean errors is quantified by bootstrap resampling of generations within bias–lever cells (20,000 replicates, 95% CIs).

Preregistered confirmatory test (not executed)

The preregistration specified a confirmatory test on a seeded random subset of 200 Block-1 problems, comparing a calibrated lever against the L0 baseline (one-sided Wilcoxon signed-rank on paired absolute errors, 95% bootstrap CI for mean ΔMAE). Because the pilot gate failed, this test was *not executed*, per the preregistered stopping rule (Section 3.3). We describe the design here for completeness and to document what the negative gate preempted; no confirmatory numbers are reported anywhere in this paper. The preregistered descriptive replication of Liu et al. (2025) (DEVIATIONS.md D1) likewise required the held-out generations and was not run; the foundation-verification attempt, which could not reproduce their absolute numbers from the public data, is documented in D1.

Results

Gate outcome: 2 of 5 biases pass

The pilot battery collected 960/960 records (903 valid; 94.1% overall parse rate, above the 90% criterion; no bias saturated above 0.95). At the bias level, endowment parses at only 85.8%, with failures concentrated in formulation 1 under the humanlike levers (L1: 1/20 parsed, L3: 10/20—empty responses). Table 1 summarizes the gate tests on the uncalibrated neutral baseline (L0). Only framing and anchoring show the human-direction effect in ≥2/3 formulations: **2/5 < 3/5, so the gate fails** and the study stops here per the preregistered rule. Power is limited by design ($n = 10$ per cell, preregistered): two formulations missed α by less than 0.01 (anchoring f2 $p = 0.057$, CRT f1 $p = 0.055$), but the null does not rest on them—the conjunction violation rate of 0.00–0.20 against a 0.85 target and the reversed WTA/WTP ordering are far from any significance boundary. A sensitivity note: with $n = 10$, a one-sided exact binomial test at $\alpha = 0.05$ needs ≥9/10 responses in the human direction (critical $p = 0.011$) and reaches 80% power only at true per-trial rates around 0.95—the gate detects only enormous effects. The two 8/10 near-misses are therefore underpowered non-detections, not evidence of absence; the structural failures survive any power critique. Under Bonferroni correction over the 15 gate tests ($\alpha = 0.0033$), framing passes 1/3, anchoring 2/3, CRT 1/3—only 1/5 biases clear the ≥2/3 rule, so the gate fails either way. (The framing f0 cell is uninformative rather than a substantive miss: all 80 responses across the four lever conditions chose option A regardless of frame—pure position bias (20/20 under L0), so the cell measured no framing effect.)

Table 1: Preregistered gate results (L0 baseline, one-sided tests, $n = 10$ per cell, $\alpha = 0.05$). A formulation passes if $p < 0.05$ in the human direction.

Bias	f0	f1	f2
Framing	$p = 0.500$	$p < 0.0001$	$p = 0.030$
Conjunction	$p = 1.000$	$p = 1.000$	$p = 0.989$
Anchoring	$p = 0.0003$	$p = 0.0001$	$p = 0.057$
Endowment	$p = 1.000$	$p = 0.988$	$p = 1.000$
CRT	$p = 0.001$	$p = 0.055$	$p = 1.000$

The failures are structural. **Conjunction:** the model essentially never commits the conjunction fallacy—violation rates 0.00, 0.00, and 0.20 across formulations against a human target of 0.85. **Endowment:** the protocol produces the reverse of the human ordering: median willingness-to-accept is *below* willingness-to-pay in all three probes (e.g., \$5.0 vs. \$50.0 in f0; \$1.0 vs. \$100.0 in f2), against the human WTA ≈ 2× WTP pattern. The elicitation—stated prices, single turn, no ownership or incentive—is a weak analogue of the ownership paradigm, so we read this as the protocol failing to produce the human ordering rather than a substantive psychological reversal: on f0 the model emits an identical \$5.00 on all 20 WTA trials, i.e., stereotyped responding rather than ownership-sensitive valuation. **CRT:** the three probes are the three classic items (bat-and-ball, machines, lily-pad), not three wordings, and performance is item-dependent: intuitive-error rates of 1.0, 0.8, and 0.0, with the model answering the lily-pad item correctly on all 10 trials. Only one item passes. The aggregate 0.60 vs. 0.60 match to the human target is an artifact of averaging heterogeneous items, not evidence of humanlikeness. Framing passes 2/3 (f0 uninformative, as above); anchoring passes 2/3 with the third just missing ($p = 0.057$), though its magnitude overshoots the human target substantially (Section 4.2).

Bias magnitudes and the calibration null

This comparison is the preregistered lever-selection analysis (Section 3.4), reported descriptively: with the gate failed, no lever is selected for confirmatory testing, and the numbers below document *why* selection would be meaningless. The preregistered strengtheners (dose-response levers, CPT fit, ablation, contamination check) were preempted by the same stopping rule and not run. Table 2 and Figure 1 show measured bias magnitudes per lever against published human effect sizes. The calibration analysis—selecting the lever minimizing mean absolute deviation from human targets—selects **L0, the neutral baseline**: every "irrationality lever" increases

Table 2: Measured bias magnitudes ([0,1] scale) per lever vs. human targets. Bold marks the closest lever per bias. The anchoring target (0.40) is our operationalization of human-typical partial adjustment, not a published effect size like the others; the endowment target (0.50) is the human ratio target 2.0 under the preregistered $\min(\text{ratio}, 4)/4$ scaling.

Bias	Target	L0	L1	L2	L3
Framing	0.50	**0.433**	0.600	0.600	0.433
Conjunction	0.85	**0.067**	0.000	0.000	0.000
Anchoring	0.40	0.919	0.933	0.913	**0.873**
Endowment	0.50	0.079	0.064	**0.151**	0.064
CRT	0.60	**0.600**	0.333	0.333	0.333

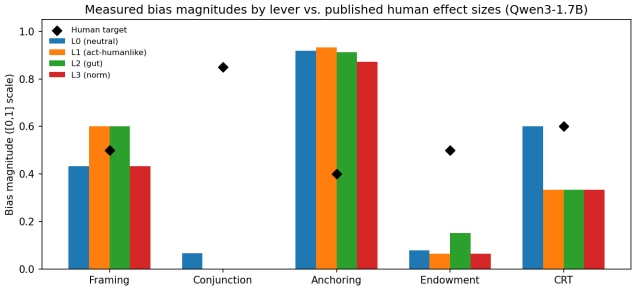


Figure 1: Measured bias magnitudes by lever against published human effect sizes (diamonds). The humanlike levers (L1–L3) move magnitudes away from, not toward, the targets on four of five biases.

miscalibration (mean |error|: L0 0.358; L1 0.437; L2 0.416; L3 0.418; bootstrap 95% CIs over resampled generations within bias–lever cells, 20,000 replicates: L0 [0.338, 0.391], L1 [0.417, 0.451], L2 [0.399, 0.434], L3 [0.404, 0.437]—no lever's interval overlaps L0's, so the degradation is not sampling noise).

The levers do not nudge the model toward human biases; they nudge it *away*. The act-humanlike and gut prompts (L1, L2) drive the conjunction violation rate from 0.067 to exactly 0.0 and cut the CRT intuitive-error rate from 0.600 to 0.333—moving the two biases that were closest to human magnitudes *farther* from them. Anchoring is the one bias where the model overshoots rather than undershoots (0.873–0.933 vs. target 0.40), and even there no lever corrects the overshoot. In short: there is nothing to calibrate. The model does not exhibit the biases in the first place, and prompting harder does not create them.

Discussion

A boundary condition, not a debunking. The literature reporting human-like biases in language models is real, and our result does not erase it—it bounds it. The positive findings come almost entirely from larger or instruction-tuned models probed with unpreregistered, often single-formulation elicitation. What we show is that under fixed, preregistered, multi-item protocols, a small model fails structurally: biases absent (conjunction), reversed in ordering (endowment), or item-dependent (reflection). "LLMs exhibit human biases" is not a scale-free property of the technology—it is a property of particular models under particular protocols. Anyone using small models as stand-ins for human decision-makers should measure the biases first rather than assume them.

Scale is the central limitation. Our model has 1.7B parameters, run at Q4_K_M quantization (a CPU-inference necessity that is itself an unexamined confound for calibration-sensitive behavior); Hagendorff et al. (2023) find intuitive-reasoning biases *emerge* with scale, and Knipper et al. (2025) find bias expression depends jointly on model size and prompt specificity. It is entirely possible that the same battery on a 70B model would pass the gate. We state this plainly: the null is scoped to the tested model and configuration, and scale is the most plausible moderator. That is itself a useful finding—it says bias expression is not scale-free.

Elicitation is the claim's boundary. Our protocols are the preregistered ones: three probe items per bias, following Pagliaro (2026)'s lesson that measurement carries large systematic uncertainty. Different prompts might surface biases we did not find. But that cuts both ways: a bias that appears only under hand-tuned elicitation is not the robust human-like property the simulation literature needs. (On CRT specifically, the heterogeneity is across the three classic items, not across wordings—item difficulty, not prompt instability.)

Prompt levers are not calibration instruments. The practically important finding is the direction of the lever effects. Act-humanlike and gut prompts—the standard tools for making models "more human" Mohr (2025)—moved measured magnitudes *away* from human targets on four of five biases. This complements Feng et al. (2026)'s warning that persona steering can redistribute rather than reduce bias: here, steering degrades even the biases that were closest to human. The cheap inference-only route to human-like simulation fails at the first step for this model class; the expensive route (fine-tuning on human choices, Binz et al. (2025)) remains the demonstrated one. A concurrent attempt at prompt-based calibration in a different domain (idea evaluation with a frontier reasoning model; Furukawa (2026)) suggests the idea is not hopeless in general—but our result says it needs biases to be present first. A caution on reading the gate's "passes": anchoring clears the directional criterion while overshooting the human magnitude more than twofold (0.87–0.93 vs. 0.40), with estimates nearly echoing the anchors themselves (17.1 vs. anchor 15; 460 vs. 450). Directional passage is not human-likeness—on magnitude the model looks more anchor-bound than humans, not human-like. Two caveats bound even this negative conclusion. First, our lever space is four fixed author-written strings, not an exhaustive search of prompt space; a different lever family might succeed where ours failed. Second, adjacent nulls point the same way as ours—Mohr (2025) reports non-detection of sunk-cost and gambler's-fallacy effects in the models tested—while Dillion et al. (2026) provides

the counterpoint that simple prompting strategies do improve AI–human alignment on moral judgment and survey items. Elicitation can help in other task families; our claim is that it did not help here, on classic decision biases, in this model.

Conclusion

We preregistered a simple, falsifiable pipeline: measure five classic decision biases in Qwen3-1.7B-Instruct under fixed protocols, require at least three to show the human-direction effect, and only then attempt prompt-based calibration to human effect sizes. The gate failed (2/5), and the calibration analysis independently confirmed the mechanism: the neutral baseline was the best-calibrated condition, and every humanlike lever made miscalibration worse. The conjunction fallacy never appears, the endowment protocol produces the reverse of the human WTA/WTP ordering, and reflection is item-dependent. Per the preregistered stopping rule, no confirmatory test was run.

The honest summary is one sentence: *you cannot calibrate irrationality that isn't there*—in this model, under these protocols, with these levers. For practitioners, the implication is to treat bias expression as a measured property of a specific model and protocol, not an assumed one—and to recognize that scale, not prompting, is the likeliest route to human-like bias where it is wanted. All materials are released for independent verification.

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Statement of AI Use

The author used the AI tool Muse (Muse Spark 1.3, Meta) for drafting, grammatical editing, and code validation. The contents are reviewed, edited, and attested for accuracy by the author.

Data and Code Availability

All code, prompts, and the one-command reproduction package are provided as supplementary material.

Competing Interests

The author declares no competing interests. No funding was received for this work.